Basis and Dimension II

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Contents

1 Dimensions of standard spaces

Recall that for a vector space V over F,

- S is a basis for V if S is linearly independent such that $V = \text{span}(S)$;
- $\dim_F(V) = |S|$.

(Sometimes we may suppress $\dim_F$ as just dim if the underlying field is clear.)

1.1 Example

1. (Vectors)

$$V = F^n$$

is a vector space over F.

$$\dim_F(F^n) = n.$$

The standard basis is

$$\{\mathbf{e_1}, \mathbf{e_2}, \ldots, \mathbf{e_n}\},$$

where

$$\{\mathbf{e_1} = (1, 0, \ldots, 0), \mathbf{e_2} = (0, 1, \ldots, 0), \ldots, \mathbf{e_n} = (0, 0, \ldots, 1)\}.$$

2. (Matrix spaces)

$$\dim_F(M_n(F)) = n^2.$$

The "standard" basis is

$$\{E_{11}, E_{12}, \ldots, E_{nn}\},$$

where E_{ij} is the matrix whose (i, j) entry is 1 and all other entries are 0.
i.e.

$$E_{11} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}, \quad E_{12} = \begin{pmatrix} 0 & 1 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}, \quad \ldots \quad E_{nn} = \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

3.
Find a basis and the dimension of the following subspace of $\mathbb{R}^4$:

$$W = \left\{(x, y, z, w) \in \mathbb{R}^4 : x + y + z + w = 0 \text{ and } -x + 2z + 3w = 0\right\}.$$

Step 1: Simplifying the equations

$$\begin{cases} x + y + z + w = 0 & (1) \\ -x + 2z + 3w = 0 & (2) \end{cases}$$

Adding (1) and (2), we obtain

$$\begin{cases} x + y + z + w = 0 & (1) \\ \quad y + 3z + 4w = 0 & (2)' \end{cases}$$

Step 2: Backward substitution
By (2)′,

$$y + 3z + 4w = 0.$$

Let

$$z = s, \qquad w = t.$$

Then

$$y = -3s - 4t.$$

By (1),

$$x + y + z + w = 0,$$

so

$$\begin{aligned} x + (-3s - 4t) + s + t &= 0 \\ x - 2s - 3t &= 0 \\ x &= 2s + 3t. \end{aligned}$$

Therefore

$$W = \{(2s + 3t, -3s - 4t, s, t) : s, t \in \mathbb{R}\}.$$

$$= \{s(2, -3, 1, 0) + t(3, -4, 0, 1) : s, t \in \mathbb{R}\}.$$

$$= \operatorname{span}\{(2, -3, 1, 0), (3, -4, 0, 1)\}.$$

As

$$B = \{(2, -3, 1, 0), (3, -4, 0, 1)\}$$

spans W, and B is linearly independent, by def, B forms a basis of W.
Therefore,

$$\dim_{\mathbb{R}}(W) = 2.$$

4. (Polynomial of degree at most n)

$$\dim_F(F[x]_{\leq n}) = n + 1.$$

The "standard" basis is

$$\{1, x, x^2, \ldots, x^n\}.$$

5. (Zero space)
Recall that

$$\{0\} = \text{span}(\varnothing),$$

where

$$\varnothing$$

denotes the empty set.
Therefore,

$$\varnothing$$

forms a basis of

$$\{0\}.$$

Hence,

$$\dim_F(\{0\}) = |\varnothing| = 0.$$

6. (Complex numbers)

$$\mathbb{C} = \{a \cdot 1 + b \cdot \sqrt{-1} : a, b \in \mathbb{R}\}.$$

Hence,

$$\dim_{\mathbb{R}}(\mathbb{C}) = 2.$$

On the other hand,

$$\mathbb{C} = \{a \cdot 1 : a \in \mathbb{C}\}.$$

Hence,

$$\dim_{\mathbb{C}}(\mathbb{C}) = 1.$$

2 Consequence of Replacement Theorem

2.1 Theorem

Theorem 3.2.1.
Let V be a vector space over F of dimension n, and let $S \subseteq V$.

1. If $|S| > n$, then S cannot be linearly independent.
2. If $|S| < n$, then S cannot span V.

(1)

Proof. Given $|S| > n$.
Suppose, for contradiction, that S is linearly independent.
Let B be a basis of V. Then

$$|B| = n.$$

Since B spans V, the Replacement Theorem implies

$$|S| \leq |B| = n,$$

which contradicts $|S| > n$.
Therefore, S cannot be linearly independent.

□

(2)

Proof. Given $|S| < n$.
Suppose, for contradiction, that S spans V.
Let B be a basis of V. Then

$$|B| = n.$$

Since B is linearly independent and S spans V, the Replacement Theorem implies

$$|B| \leq |S|.$$

Hence

$$n = |B| \leq |S|,$$

which contradicts $|S| < n$.
Therefore, S cannot span V □

Theorem 3.2.2. Let V and W be finite-dimensional vector spaces.
If

1. $W \subseteq V$,
2. $\dim_F(W) = \dim_F(V)$,

then $W = V$.

Proof. Given

$$W \subseteq V$$

and

$$\dim_F(W) = \dim_F(V) = n.$$

We WTS that $V \subseteq W$.
Suppose not. Then there exists

$$\mathbf{v} \in V$$

such that

$$\mathbf{v} \notin W.$$

Let

$$S = \{\mathbf{w_1}, \ldots, \mathbf{w_n}\}$$

be a basis of W.
Since

$$W = \text{span}(S),$$

we have

$$\mathbf{v} \notin \text{span}(S).$$

Therefore,

$$S' = \{\mathbf{w_1}, \ldots, \mathbf{w_n}, \mathbf{v}\}$$

is linearly independent in V.
Since

$$\dim_F(V) = n,$$

Theorem 3.2.1 implies that any linearly independent subset of V contains at most n vectors.
However,

$$|S'| = n + 1,$$

a contradiction.
Hence

$$V \subseteq W.$$

Together with

$$W \subseteq V,$$

we conclude that

$$W = V.$$

□

Theorem 3.2.3. Let V be a vector space of dimension n. Then any n linearly independent vectors form a basis of V.
(In literature, we say that any maximal linearly independent subset forms a basis.)

Proof. Let S be a linearly independent subset of V with

$$|S| = n.$$

We WTS that

$$\operatorname{span}(S) = V.$$

Let

$$W = \operatorname{span}(S).$$

Then W is a subspace of V, so

$$W \subseteq V.$$

Since S spans W and is linearly independent, S is a basis of W.
Therefore,

$$\dim_F(W) = |S| = n.$$

Since

$$\dim_F(V) = n,$$

we have

$$\dim_F(W) = \dim_F(V).$$

By Theorem 3.2.2,

$$W = V.$$

Hence

$$\operatorname{span}(S) = V.$$

Therefore S is a basis of V.

□

2.2 Example

1. $S_1 = \{(1,0,1),(1,1,0),(2,3,1),(-1,2,3)\}$ cannot be linearly independent.

$$S_1 = \{(1,0,1),(1,1,0),(2,3,1),(-1,2,3)\}.$$

Since

$$\dim_{\mathbb{R}}(\mathbb{R}^3) = 3,$$

Theorem 3.2.1 implies that any linearly independent subset of $\mathbb{R}^3$ contains at most 3 vectors.
However,

$$|S_1| = 4.$$

Therefore S_1 cannot be linearly independent.

2. $S_2 = \{(1,0,1),(1,1,0)\}$ cannot span $\mathbb{R}^3$.

$$S_2 = \{(1,0,1),(1,1,0)\}.$$

Since

$$\dim_{\mathbb{R}}(\mathbb{R}^3) = 3,$$

Theorem 3.2.1 implies that any spanning subset of$\mathbb{R}^3$ must contain at least 3 vectors.
However,

$$|S_2| = 2.$$

Therefore S_2 cannot span $\mathbb{R}^3$.

3. The subset $S = \{(1,0,1),(1,1,0),(2,3,1)\}$ forms a basis of $\mathbb{R}^3$.
(Check.) S is linearly independent.
Since

$$\dim_{\mathbb{R}}(\mathbb{R}^3) = 3 = |S|,$$

S is a maximal linearly independent set. Therefore S is a basis of $\mathbb{R}^3$.
Remark: This avoids the tedious check of spanning.

4. Let $V = M_2(\mathbb{R})$ and consider the subset

$$S = \left\{ A_1 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix},\ A_2 = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix},\ A_3 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},\ A_4 = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix},\ A_5 = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix},\ A_6 = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} \right\}.$$

(a) Prove that $\operatorname{span}(S) = V$

Proof. Let

$$S' = \{A_1, A_2, A_3, A_4\}.$$

Suppose

$$a_1A_1 + a_2A_2 + a_3A_3 + a_4A_4 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

By comparing the entries, we obtain

$$\begin{cases} a_1 + a_2 + a_3 = 0, \\ a_1 + a_4 = 0, \\ a_2 = 0, \\ a_3 + a_4 = 0. \end{cases}$$

Hence

$$a_1 = a_2 = a_3 = a_4 = 0.$$

Therefore S' is linearly independent.
Since

$$|S'| = 4 = \dim_{\mathbb{R}}(M_2(\mathbb{R})),$$

Theorem 3.2.3 implies that S' is a basis of V.
Thus

$$V = \operatorname{span}(S').$$

Since

$$S' \subseteq S,$$

we conclude that

$$\operatorname{span}(S) = V.$$

□

(b) Find a subset of S that is a basis of V
By part (a),

$$\{A_1, A_2, A_3, A_4\}$$

is a basis of V.

(c) Find a subset of S with 4 elements which is not a basis of V
Observe that

$$A_1 + A_6 = A_2 + A_4 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Therefore

$$A_1 - A_2 - A_4 + A_6 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Since the coefficients are not all zero,

$$\{A_1, A_2, A_4, A_6\}$$

is linearly dependent.
Hence

$$\{A_1, A_2, A_4, A_6\}$$

is not a basis of V.

3 Sifiting and proof of Replacement Theorem

3.1 Theorem

Theorem 3.3.1. A subset

$$S = \{\mathbf{v_1}, \mathbf{v_2}, \dots, \mathbf{v_m}\}$$

of V is linearly independent if and only if for each $r = 1, \dots, m$, v_r is not a linear combination of $\mathbf{v_1}, \mathbf{v_2}, \dots, \mathbf{v_{r-1}}$. In particular, this requires $\mathbf{v_1} \neq \mathbf{0}$.

Proof. This is just Theorem 2.2.1. □

Theorem 3.3.2. Let

$$S = \{\mathbf{v_1}, \mathbf{v_2}, \dots, \mathbf{v_m}, \mathbf{w}\}.$$

If w is a linear combination of $\mathbf{v_1}, \mathbf{v_2}, \dots, \mathbf{v_m}$, then

$$\text{span}(S) = \text{span}\{\mathbf{v_1}, \mathbf{v_2}, \dots, \mathbf{v_m}\}.$$

Proof. ($\subseteq$)
Given

$$\mathbf{w} = a_1\mathbf{v_1} + a_2\mathbf{v_2} + \cdots + a_m\mathbf{v_m}$$

for some $a_i \in F$.
We first show

$$\text{span}(S) \subseteq \text{span}\{\mathbf{v_1}, \mathbf{v_2}, \dots, \mathbf{v_m}\}.$$

Let $\mathbf{v} \in \text{span}(S)$. Then

$$\begin{aligned}
\mathbf{v} &= b_1\mathbf{v_1} + b_2\mathbf{v_2} + \cdots + b_m\mathbf{v_m} + c\mathbf{w} \\
&= b_1\mathbf{v_1} + b_2\mathbf{v_2} + \cdots + b_m\mathbf{v_m} + c(a_1\mathbf{v_1} + \cdots + a_m\mathbf{v_m}) \\
&= (b_1 + ca_1)\mathbf{v_1} + (b_2 + ca_2)\mathbf{v_2} + \cdots + (b_m + ca_m)\mathbf{v_m}.
\end{aligned}$$

Hence

$$\mathbf{v} \in \text{span}\{\mathbf{v_1}, \mathbf{v_2}, \dots, \mathbf{v_m}\},$$

so

$$\text{span}(S) \subseteq \text{span}\{\mathbf{v_1}, \mathbf{v_2}, \dots, \mathbf{v_m}\}.$$

($\subseteq$)
Conversely, let

$$\mathbf{v} \in \text{span}\{\mathbf{v_1}, \mathbf{v_2}, \dots, \mathbf{v_m}\}.$$

Then

$$\begin{aligned}
\mathbf{v} &= d_1\mathbf{v_1} + d_2\mathbf{v_2} + \cdots + d_m\mathbf{v_m} \\
&= \underbrace{d_1\mathbf{v_1} + d_2\mathbf{v_2} + \cdots + d_m\mathbf{v_m} + 0 \cdot \mathbf{w}}_{\text{linear combination of } S}.
\end{aligned}$$

Hence

$$\mathbf{v} \in \operatorname{span}(S),$$

which implies

$$\operatorname{span}\{\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_m}\} \subseteq \operatorname{span}(S).$$

Therefore,

$$\operatorname{span}(S) = \operatorname{span}\{\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_m}\}.$$

□

Theorem 3.3.3. (Sifiting method) Let

$$S = \{\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_m}\}$$

be a subset of V. Then there exists a subset

$$S' \subseteq S$$

such that

1. S' is linearly independent; and
2. $\operatorname{span}(S') = \operatorname{span}(S)$.

Proof. For each r, check whether $\mathbf{v_r}$ is a linear combination of the previous vectors

$$\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_{r-1}}.$$

1. Keep $\mathbf{v_r}$ in S if $\mathbf{v_r}$ is *not* a linear combination of $\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_{r-1}}$ (not redundant).
2. Remove $\mathbf{v_r}$ from S if $\mathbf{v_r}$ *is* a linear combination of $\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_{r-1}}$ (redundant).

After this process, we obtain a subset S' in which no element is a linear combination of the preceding members. Therefore, by Theorem 3.3.1, S' is linearly independent. Moreover, by Theorem 3.3.2, removing redundant members from S does not change its linear span. Hence, we have $\operatorname{span}(S') = \operatorname{span}(S)$.

□

Theorem 3.3.4. (Basis extension theorem)
Let V be a finite-dimensional vector space, and let S be a linearly independent subset of V. Then S can be extended into a basis of V.

Proof. Let
$$S = \{\mathbf{v_1}, \dots, \mathbf{v_m}\}$$
be a linearly independent subset of V.
Let
$$B = \{b_1, \dots, b_n\}$$
be any basis of V.
Consider
$$S \cup B = \{\mathbf{v_1}, \dots, \mathbf{v_m}, \mathbf{b_1}, \dots, \mathbf{b_n}\}.$$
Apply the Sifting Method to $S \cup B$.
Since S is linearly independent, none of the vectors $\mathbf{v_1}, \dots, \mathbf{v_m}$ will be removed during the sifting process.
Hence we obtain
$$S \cup B', \qquad \text{where } B' \subseteq B.$$
Notice that

1. $S \cup B'$ is linearly independent;

2. $\text{span}(S \cup B') = \text{span}(S \cup B)$.

Since B is a basis of V,
$$\text{span}(S \cup B) = V.$$
Therefore,
$$\text{span}(S \cup B') = V.$$
Hence $S \cup B'$ is both linearly independent and spans V. Therefore,
$$S \cup B'$$
forms a basis of V, which is an extension of S. □

3.2 Proof of Exchange/Replacement Theorem

This is the proof of Replacement Theorem (Theorem 2.5.1) which says that if

- $S_1 = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a spanning set of V.
- $S_2 = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m\}$ is linearly independent.

Then

$$m \leq n.$$

Proof. Idea: Insert $\mathbf{w}_j$ one-by-one into S_1, then apply the Sifting Method.
Consider

$$B_1 = \{\mathbf{w}_1, \mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\} = \{\mathbf{w}_1\} \cup S_1.$$

Applying the Sifting Method to B_1, we must remove some $\mathbf{v}_i$'s. Hence we obtain

$$B_1' = \{\mathbf{w}_1\} \cup S_1', \text{ where } |S_1'| < |S_1|.$$

Moreover,

$$B_1'$$

remains a spanning set of V.
Now consider

$$B_2 = \{\mathbf{w}_2, \mathbf{w}_1\} \cup S_1'.$$

Applying the Sifting Method again, none of the $\mathbf{w}_i$'s will be removed because

$$S_2 = \{\mathbf{w}_1, \ldots, \mathbf{w}_m\}$$

is linearly independent.
Hence we obtain

$$B_2' = \{\mathbf{w}_2, \mathbf{w}_1\} \cup S_1'', \text{ where } |S_1''| < |S_1'|.$$

Continue this process. Finally we obtain

$$B_m' = \{\mathbf{w}_m, \mathbf{w}_{m-1}, \ldots, \mathbf{w}_1\} \cup S_1^{(m)},$$

where

$$0 \leq |S_1^{(m)}| < |S_1^{(m-1)}| < \cdots < |S_1''| < |S_1'| < |S_1| = n.$$

Thus there are at least m distinct integers strictly less than n.
Therefore, we must have

$$m \leq n.$$

□

4 Direct sum of subspaces

4.1 Definition

Definition 3.4.1. (Sum of subspaces)
Let W_1, W_2 be subspaces of a vector space V. Then define

$$W_1 + W_2 := \{\mathbf{w_1} + \mathbf{w_2} : \mathbf{w_1} \in W_1 \text{ and } \mathbf{w_2} \in W_2\}$$

Definition 3.4.2. (Direct Sum)
We write $W_1 + W_2 = W_1 \bigoplus W_2$ if $W_1 \cap W_2 = \{\mathbf{0}\}$. In this case, we call it the *direct sum* (instead of just sum) of W_1 and W_2.

4.2 Theorem

Theorem 3.4.1.
Let $W_1 + W_2$ be a subspaces of V. Indeed, this is smallest subspace containing both W_1 and W_2

Proof. We verify the three subspace conditions.

$\boxed{\text{Non-empty}}$
Since W_1 and W_2 are subspaces,

$$\mathbf{0} \in W_1, \qquad \mathbf{0} \in W_2.$$

Hence

$$\mathbf{0} = \mathbf{0} + \mathbf{0} \in W_1 + W_2. \implies W_1 + W_2 \text{ is non-empty.}$$

$\boxed{\text{Closed under addition}}$
Let

$$\mathbf{a}, \mathbf{b} \in W_1 + W_2.$$

Then

$$\begin{cases} \mathbf{a} = \mathbf{w}_1 + \mathbf{w}_2 & \text{for some } \mathbf{w}_1 \in W_1 \text{ and } \mathbf{w}_2 \in W_2. \\ \mathbf{b} = \mathbf{w}_1' + \mathbf{w}_2' & \text{for some } \mathbf{w}_1' \in W_1 \text{ and } \mathbf{w}_2' \in W_2. \end{cases}$$

Therefore,

$$\mathbf{a} + \mathbf{b} = (\mathbf{w}_1 + \mathbf{w}_1') + (\mathbf{w}_2 + \mathbf{w}_2').$$

Since W_1 and W_2 are subspaces,

$$\mathbf{w}_1 + \mathbf{w}_1' \in W_1, \qquad \mathbf{w}_2 + \mathbf{w}_2' \in W_2.$$

Hence

$$\mathbf{a}+\mathbf{b} \in W_1 + W_2. \implies W_1 + W_2 \text{ is Closed under addition}$$

$\boxed{\text{Closed under scalar multiplication}}$

Let

$$\mathbf{a} \in W_1 + W_2, \text{ and } c \text{ is a scalar}$$

Then

$$\begin{cases} \mathbf{a} = \mathbf{w}_1 + \mathbf{w}_2 \text{ for some } \mathbf{w_1} \in W_1 \text{ and } \mathbf{w_2} \in W_2 \\ c\mathbf{a} = c(\mathbf{w}_1 + \mathbf{w}_2) = c\mathbf{w_1} + c\mathbf{w_2} \text{ for some } \mathbf{w_1} \in W_1 \text{ and } \mathbf{w_2} \in W_2 \end{cases}$$

Since

$$c\mathbf{w}_1 \in W_1, \qquad c\mathbf{w}_2 \in W_2,$$

we conclude that

$$c\mathbf{a} \in W_1 + W_2.$$

Therefore, $W_1 + W_2$ is a subspace of V. □

Theorem 3.4.2.
Let V be a finite-dimensional vector space, and let W_1, W_2 be subspaces of V. Then

$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2).$$

Proof. $\boxed{\text{Idea. Skill for dimension formula: Start with the “smallest” subspace}}$

$$\boxed{W_1 \cap W_2 \subseteq W_1,\ W_2 \subseteq W_1 + W_2.}$$

Let

$$n = \dim(W_1 \cap W_2),$$

and let

$$B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$$

be a basis of $W_1 \cap W_2$.
Since $B \subseteq W_1$ and B is linearly independent, by the Basis Extension Theorem, we can extend B to a basis of W_1:

$$B_1 = \{\mathbf{b}_1, \dots, \mathbf{b}_n, \mathbf{s}_1, \dots, \mathbf{s}_p\}, \text{ a basis of } W_1$$

Similarly, since $B \subseteq W_2$ and B is linearly independent, we may extend B to a basis of W_2:

$$B_2 = \{\mathbf{b}_1, \dots, \mathbf{b}_n, \mathbf{t}_1, \dots, \mathbf{t}_q\}, \text{ a basis of } W_2$$

$\boxed{\text{BIG CLAIM}}$

Let

$$S = \{\mathbf{b}_1, \ldots, \mathbf{b}_n, \mathbf{s}_1, \ldots, \mathbf{s}_p, \mathbf{t}_1, \ldots, \mathbf{t}_q\}.$$

We claim that S forms a basis of $W_1 + W_2$.

Proof of the claim. We verify the three subspace conditions:

(1) $\boxed{S \text{ spans } W_1 + W_2}$

Let

$$\mathbf{w} \in W_1 + W_2.$$

Then there $\exists$ $\mathbf{w}_1 \in W_1$ and $\mathbf{w}_2 \in W_2$ s.t.

$$\mathbf{w} = \mathbf{w}_1 + \mathbf{w}_2 \in W_1 + W_2$$

Since B_1 is a basis of W_1,

$$\mathbf{w}_1 = \sum_{i=1}^{n} \alpha_i \mathbf{b}_i + \sum_{j=1}^{p} \beta_j \mathbf{s}_j.$$

And B_2 is a basis of W_2,

$$\mathbf{w}_2 = \sum_{i=1}^{n} \alpha_i' \mathbf{b}_i + \sum_{k=1}^{q} \gamma_k \mathbf{t}_k.$$

So

$$\begin{aligned}\mathbf{w} &= \mathbf{w}_1 + \mathbf{w}_2 \\ &= \sum_{i=1}^{n} (\alpha_i + \alpha_i')\mathbf{b}_i + \sum_{j=1}^{p} \beta_j \mathbf{s}_j + \sum_{k=1}^{q} \gamma_k \mathbf{t}_k.\end{aligned}$$

Therefore,

$$\mathbf{w} \in \mathrm{span}(S).$$

Hence S spans $W_1 + W_2$.

(2) $\boxed{S \text{ is linearly independent.}}$

Suppose $\exists$ scalars α_i, β_j, γ_k s.t.

$$\sum_{i=1}^{n} \alpha_i \mathbf{b}_i + \sum_{j=1}^{p} \beta_j \mathbf{s}_j + \sum_{i=1}^{q} \gamma_k \mathbf{t}_k = \mathbf{0}.$$

Rearranging,

$$\sum_{i=1}^{n} \alpha_i \mathbf{b}_i + \sum_{j=1}^{p} \beta_j \mathbf{s}_j = -\sum_{k=1}^{q} \gamma_k \mathbf{t}_k.$$

Let $\mathbf{v}$ be left-hand side. Undoubtedly, $\mathbf{v}$ equals right-hand side.
Clearly, $\mathbf{v}$ belongs to both W_1 and W_2. Hence both sides belong to $W_1 \cap W_2$.
As

$$B = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$$

is a basis of $W_1 \cap W_2$, $\mathbf{v}$ can be written as

$$\mathbf{v} = \sum_{i=1}^{n} \sigma_i \mathbf{b}_i \implies \mathbf{0} = \sum_{i=1}^{n} \sigma_i \mathbf{b}_i + \sum_{k=1}^{q} \gamma_k \mathbf{t_k}$$

As $B_2 = \{\mathbf{b}_1, \ldots \mathbf{b}_n,, \mathbf{t}_1, \ldots \mathbf{t}_q\}$ is linearly independent, this forces $\boxed{\sigma_i = 0 \ \forall i}$, $\boxed{\gamma_k = 0 \ \forall k}$
Then the rearranging part above can be reduced to

$$\sum_{i=1}^{n} \alpha_i \mathbf{b}_i + \sum_{j=1}^{p} \beta_j \mathbf{s}_j = \mathbf{0}.$$

As $B_1 = \{\mathbf{b}_1, \ldots \mathbf{b}_n,, \mathbf{s}_1, \ldots \mathbf{s}_p\}$ is linearly independent, this forces $\boxed{\alpha_i = 0 \ \forall i}$, $\boxed{\beta_j = 0 \ \forall j}$
So S is linearly independent.

□

Hence S is a basis of $W_1 + W_2$,

$$\begin{aligned} \dim(W_1 + W_2) &= |S| \\ &= n + p + q \\ &= (n + p) + (n + q) - n \\ &= \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2). \end{aligned}$$

□

Theorem 3.4.3.
If W_1, W_2 are finite-dimensional subspaces of V with $W_1 \cap W_2 = \{\mathbf{0}\}$, then

$$\dim(W_1 \oplus W_2) = \dim(W_1) + \dim(W_2).$$

Proof. Since

$$W_1 \oplus W_2 = W_1 + W_2,$$

By Theorem 3.4.2, we have

$$\begin{aligned} \dim(W_1 \oplus W_2) &= \dim(W_1 + W_2) \\ &= \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2) \\ &= \dim(W_1) + \dim(W_2) - 0 \\ &= \dim(W_1) + \dim(W_2). \end{aligned}$$

□

Theorem 3.4.4.
Let V be a finite-dimensional vector space and let W be a subspace of V. Then there exists another subspace W' of V such that $V = W \oplus W'$.

In this case, we call W' A *complementary subspace* of W in V.

Proof. Let

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$$

be a basis of W.
By the Basis Extension Theorem, we may extend S to a basis of V, and we say that

$$S' = \{\underbrace{\mathbf{v}_1, \dots, \mathbf{v}_k}_{\text{span W}}, \mathbf{s}_1, \dots, \mathbf{s}_n\}.$$

Define

$$W' = \operatorname{span}\{\mathbf{s}_1, \dots, \mathbf{s}_n\}.$$

Let $V = W \bigoplus W'$, we claim that

$$(1)\ V \subseteq W + W', \qquad (2)\ W \cap W' = \{\mathbf{0}\}.$$

proof of (1). Let

$$\mathbf{v} \in V.$$

As S' is a basis of V,

$$\mathbf{v} = \underbrace{\sum_{i=1}^{k} \alpha_i \mathbf{v}_i}_{\in W} + \underbrace{\sum_{j=1}^{n} \beta_j \mathbf{s}_j}_{\in W'} \in W + W'$$

□

proof of (2). Let

$$\mathbf{w} \in W \cap W'.$$

Since both $\mathbf{w} \in W$ and $\mathbf{w} \in W'$,

$$\mathbf{w} = \sum_{i=1}^{k} \alpha_i \mathbf{v}_i = \sum_{j=1}^{n} \beta_j \mathbf{s}_j.$$

Hence,

$$\sum_{i=1}^{k} \alpha_i \mathbf{v}_i - \sum_{j=1}^{n} \beta_j \mathbf{s}_j = \mathbf{0}.$$

Since S' is linearly independent,

$$\alpha_i = \beta_j = 0, \quad \forall i, j$$

Thus, $\mathbf{w} = \mathbf{0}, \implies W \cap W' = \{\mathbf{0}\}$.

$\square$

$\square$

4.3 Question

Question: Is true that, $\dim(W_1 + W_2 + W_3) = \dim(W_1) + \dim(W_2) + \dim(W_3) - \dim(W_1 \cap W_2) - \dim(W_2 \cap W_3) - \dim(W_3 \cap W_1) + \dim(W_1 \cap W_2 \cap W_3)$?
Answer: False.

4.4 Example

4.4.1 Direct sum of symmetric and skew-symmetric matrices

Consider the vector space $V = M_n(\mathbb{C})$ and two of its subspaces :

- $W_1 = \{A \in M_n(\mathbb{C}) \mid A^t = A\}$
- $W_2 = \{A \in M_n(\mathbb{C}) \mid A^t = -A\}$

Prove that $V = W_1 \oplus W_2$.

Proof. We need to show

(1) $V = W_1 + W_2$
i.e. any square matrix is a sum of a symmetric and a skew-symmetric matrix

(2) $W_1 \cap W_2 = \{\mathbf{0}\}$ (zero matrix)

proof of (1). Let

$$A \in M_n(\mathbb{C}).$$

Observe that

$$A = \frac{1}{2}(A + A^t) + \frac{1}{2}(A - A^t).$$

Let

$$B = \frac{1}{2}(A + A^t), \qquad C = \frac{1}{2}(A - A^t).$$

Then

$$B^t = \frac{1}{2}(A + A^t)^t = \frac{1}{2}(A^t + A) = B,$$

so B is symmetric.

Similarly,

$$C^t = \frac{1}{2}(A - A^t)^t = \frac{1}{2}(A^t - A) = -C,$$

so C is skew-symmetric

Therefore,

$$A = B + C \in W_1 + W_2,$$

which implies

$$V = W_1 + W_2.$$

$\square$

proof of (2). Let

$$A \in W_1 \cap W_2.$$

Then

$$A^t = A, \quad A^t = -A.$$

Hence,

$$A = -A,$$

so

$$2A = \mathbf{0}.$$

Since the underlying field is $\mathbb{C}$,

$$A = \mathbf{0}.$$

Therefore,

$$W_1 \cap W_2 = \{\mathbf{0}\}.$$

$\square$

Combining (1) and (2), we conclude that

$$V = W_1 \oplus W_2.$$

$\square$